

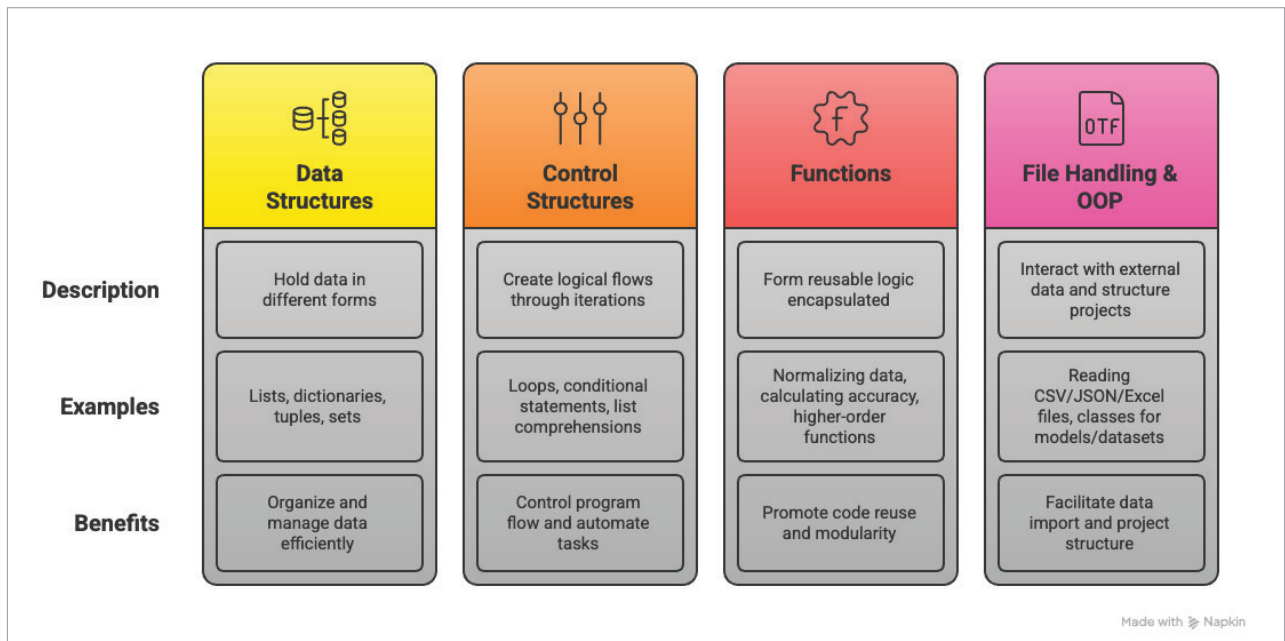


Figure 2: Python programming components for machine learning

on top of it, providing high-level representations of statistical plots. Such tools help the developer in making histograms, box plots, scatter plots, and correlation matrices to present data in an effective manner.

scikit-learn: scikit-learn is one of the most popular libraries in classical machine learning. It provides a consistent interface for model training, feature selection, data splitting, and evaluation of the model performance. From regression and classification, to clustering and dimensionality reduction, scikit-learn aims to ease implementation with clean, simple, and consistent APIs.

TensorFlow and Keras: TensorFlow is a widely used library developed by Google for numerical computation and deep learning. It can build highly sophisticated neural network architectures and supports training as well as inferences on CPU, GPU, or TPU. Slightly advanced in terms of the UI, Keras comes with TensorFlow and allows for better prototyping and experimentation.

PyTorch: PyTorch's dynamic computation graphs and flexibility were developed by the AI Research Lab of Facebook, making it a highly acclaimed library, especially in academia and research. However, PyTorch can also be deployed in production systems, where it is supported by additional libraries like TorchServe.

Key steps in a machine learning pipeline

Machine learning modelling is complicated and structured. Some of the critical steps in a typical machine learning pipeline are outlined below, illustrated with the relevant code examples.

Data collection

The first step in any machine learning pipeline is data collection. Machine learning models depend largely on data to acquire their patterns and make predictions. Without high-quality, representative data, predictions from the model could be wrong or irrelevant. Data is collected from various sources — structured databases, raw files like CSV or Excel, or any API to fetch real-time data.

After the data is collected, it is important to check for its quality and relevance to the task at hand. In other words, if you are working on a classification task to predict customer churn, the dataset must include relevant features like customer behaviour, demographics, and purchase history. Python tools generally used for loading and exploring the datasets are Pandas and NumPy, which have the best support for operating and manipulating structured data.

```
import pandas as pd
# Reading data from a CSV file
df = pd.read_csv('data.csv')

# Displaying the first few rows of the data
print(df.head())
```

Data preprocessing

Raw data is often incomplete or inconsistent, and must be cleaned and prepared for training. Incomplete data can also lead to predictions that are biased. Hence, the first concern is missing data. Data with missing values can be deleted or an estimated value can be added. The second concern